$$\begin{cases} (1-3j)x + (2j+1)y = 5 \\ (9-4j)x - (2+j)y = 3+11j \end{cases}$$

$$\Delta = \begin{vmatrix} 1-3j & 1+2j \\ 9-4j & -(2+j) \end{vmatrix} = -(1-3j)\cdot(2+j) - (1+2j)(9-4j)$$

$$= (-1+3j)(2+j) + (-1-2j)(9-4j) =$$

$$= -2 - j + 6j - 3 - 9 + 4j - 18j - 8 =$$

$$= -22 - 9j \neq 0$$

$$\Delta x = \begin{vmatrix} 5 & 1+2j \\ 3+11j & -2-j \end{vmatrix} = -10 - 5j + (3+11j)(-1-2j) =$$

$$= -10 - 5j - 3 - 6j - 11j + 22 = 9 - 22j$$

$$\Delta y = \begin{vmatrix} 1-3j & 5 \\ 9-4j & 3+11j \end{vmatrix} = (1-3j)(3+11j) - 5(9-4j) =$$

$$= 3 + 11j - 9j + 33 - 45 + 20 = -9 + 9 + 22j$$

$$x = \frac{9-22j}{-22-9j} = j \qquad y = \frac{-9+22j}{-22-9j} = -j$$